Azure Private Endpoints and DNS

Day 02 | AZURE / BLOB PRIVATE ENDPOINT + DNS

Use this guide to investigate a specific failure, apply a cause-specific change and verify recovery. Start with evidence from the affected workload. The examples require adaptation to your environment.

How the failure happens

The application asks DNS for the usual storage hostname. Its resolver must follow the private-link alias to an A record containing the private endpoint address. The endpoint existing in a subnet does not make every resolver use the associated private DNS zone.

The Scenario A Blob private endpoint exists, but an app in the VNet still resolves the storage hostname publicly.

What Is Missing? Private connectivity and private name resolution are separate requirements.

The Boundary This example is Azure Blob. Other subresources use different private DNS zones.

Before you start

An approved Azure Blob private endpoint; the correct private zone and record; known client VNet; resolver access; Blob data permissions.

Diagnostic checks

```
nslookup <account>.blob.core.windows.net

# Run from the actual app network
# Inspect the CNAME and final A record
```

Check The Client Path: Confirm which DNS server the app uses and whether it can resolve the linked private zone.

Check The Endpoint: Confirm approval, private IP, DNS zone group and VNet access. Peering alone does not link private DNS zones.

Apply the cause specific fix

Fix The Confirmed Cause

```
# Terraform: existing zone + client VNet
resource "azurerm_private_dns_zone_virtual_network_link" "app" {
  name                  = "app-link"
  resource_group_name   = var.dns_rg
  private_dns_zone_name = var.blob_zone
  virtual_network_id    = var.app_vnet_id
  registration_enabled  = false
}
```

For This Scenario: Link the existing privatelink.blob.core.windows.net zone to the app VNet. Ensure the endpoint A record exists.

Custom Dns?: Configure the resolver and forwarding path too. A VNet link alone does not fix every custom DNS topology.

Validate the result

```
nslookup <account>.blob.core.windows.net

# Then test HTTPS + authorized Blob access
```

Expected: The final address matches the private endpoint IP from the app network.

Still Failing?: Then check routes, firewall rules and data-plane permissions separately.

Practice exercise

Query from inside and outside the intended network. Document the CNAME chain and final address. Change only the confirmed DNS defect, then test authorized access using the normal hostname.

Implementation notes

HCL fragment requires the azurerm provider, existing DNS zone, its A record/zone group and variables. It does not provision a private endpoint. Use the normal service FQDN for TLS, not the private IP. Do not disable public access until required client paths are tested. Custom/on-prem DNS needs appropriate forwarding or Azure DNS Private Resolver.

Interview question

Why can a VM resolve a private IP while a hosted pipeline agent resolves a public IP? Compare their network and resolver paths.

Record what changed

- The symptom and affected user operation.
- Evidence supporting the root cause and alternatives ruled out.
- The changed configuration and the version or identity involved.
- Positive and negative tests, plus any remaining risks.

Example status

Examples are educational templates or fragments. They were not deployed against a live cloud account or production cluster. A successful syntax check is not proof of runtime compatibility. Test adapted configurations in an isolated environment first.

Official references

Official documentation 1